

Emergency / Non-Emergency Work Order Classification

TX Work Order — Feature Plan & Emergency Definitions • July 9, 2026 • Source: Chana / Nathan (Asana: IT Development and Tickets)

1. Goal

Automatically classify every incoming work order as **Emergency** or **Non-Emergency** at intake, using the criteria defined below. Staff can always review and override. In a later phase, the same classification logic will power an **AI Receptionist** that recognizes emergencies during tenant calls.

2. What Is an Emergency? (Official Definition)

"An emergency maintenance issue is any condition that threatens the safety of occupants, causes active damage to the property, renders the property uninhabitable, or creates a significant risk of further damage if not addressed immediately. Emergencies will be responded to as quickly as possible, including after-hours when necessary. Routine maintenance requests will be addressed during normal business hours."

For Houston property management, the most common after-hours emergencies are: **active water leaks, sewer backups, gas leaks, electrical hazards, and HVAC failures during extreme heat.**

3. Emergency Categories

Category	Qualifying Conditions
Water	Active water leaks causing damage • Burst water lines • Overflowing toilets that cannot be stopped • Flooding from plumbing failures • Sewer backups inside the property
Electrical	Burning smell from outlets or electrical panels • Sparking outlets, switches, or wiring • Electrical fire hazards • Complete power outage affecting only the property (not utility-wide)
HVAC	Complete loss of A/C when indoor temperatures create health risks (Houston summers) • Complete loss of heat during freezing temperatures • Carbon monoxide concerns from heating equipment • Refrigerant leaks creating safety concerns
Fire & Smoke	Active fire • Smoke inside the property • Smoke detector issues indicating possible fire • Burnt electrical components
Gas	Gas odor inside or around the property • Suspected gas leaks • Damaged gas lines
Security	Broken exterior doors that cannot be secured • Broken locks preventing secure occupancy • Broken windows creating immediate security risk • Forced entry or burglary damage
Structural	Ceiling or roof collapse • Major storm damage exposing the interior • Significant foundation or structural failures • Large trees fallen on the structure

Health & Safety	Carbon monoxide alarm activation • Significant mold caused by active flooding • No functioning toilet in a single-bathroom home • Any situation creating immediate health hazards
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4. NOT Emergencies (Urgent Maintenance, Normal Business Hours)

Issue	Why It Can Wait
Dripping faucets	No active damage
Running toilets (unless overflowing)	No active damage
Garbage disposal not working	Convenience issue
One appliance not working	Convenience issue
One sink clogged	Other fixtures available
Minor roof leaks during dry weather	No immediate damage risk
Pest sightings	Routine treatment
Cosmetic issues	No safety or damage risk
A/C working but not cooling to tenant preference	System is functional
Loss of hot water	Urgent, but typically not an emergency

5. What Already Exists in the System

Good news: the system already has most of the foundation in place.

- **Emergency flag exists:** `work_orders.is_emergency` — three states: unclassified (null), Emergency, Non-emergency. Staff set it today with a manual dropdown on the work order page.
- **Emergency drives tasks:** changing the flag automatically regenerates the work order's task checklist from emergency-specific task templates.
- **An AI review pipeline exists** that classifies a work order's issue type and vendor category with a confidence score — but today it only runs when a staff member requests it on-demand in the UI. The scheduled 10-minute PropertyWare import does **not** trigger it automatically (Phase 1 fixes this).
- **Emergency badges already show** in search results, the vendor portal (red highlight), and work order cards.
- **What's missing:** nothing sets `is_emergency` automatically — every work order arrives unclassified until a staff member manually toggles it. There is also no alert/notification when something is an emergency.

6. Proposed Solution

Phase 1 — AI Emergency Classification at Intake

- **Wire the AI review into the scheduled import:** the 10-minute PropertyWare sync triggers the AI review job for each **newly created** work order only (today that job only runs on-demand from the UI). Existing work orders are never re-classified.
- Teach the existing AI work order agent Chana's criteria (Sections 2–4 above) and have it return: **Emergency / Non-Emergency**, the emergency **category** (Water, Gas, etc.), a **confidence score**, and a one-line **reason**.
- Store the result alongside the existing AI recommendation for each work order.
- **Every new work order is labeled automatically** — Emergency or Non-Emergency, no review step (per IT lead). When details are missing the AI judges by the most likely Houston-rental reading of the criteria, and genuinely ambiguous safety/damage cases lean Emergency (missing a real emergency is worse than over-flagging one). A staff member's manual setting always wins and is never overwritten.
- Add a keyword-based fallback (gas, flood, sparks, fire, sewage...) so classification still works if the AI service is down.

Phase 2 — Transparency UI

- The work order's Recommendation panel shows the AI's label with its **category, confidence, and one-line reason** — purely informational, no action required.
- Shows **who/what** classified it (AI vs. staff) for accountability. A wrong label is corrected with the existing Emergency dropdown on the work order page.
- An Emergency filter on the dashboard (Emergency / Non-emergency / Unclassified) so emergencies are impossible to miss.

Phase 3 — Emergency Alerts

- **In-app alert to all staff:** uses the existing notification bell. Staff already see all system notifications; tenants and vendors are automatically excluded by the existing role filters. No new infrastructure needed.
- **SMS to the assigned Work Order Coordinator (WOC):** every imported work order is already assigned a WOC, and Twilio is already integrated — the emergency alert texts the WOC responsible for that work order.
- Slack and email are deliberately left out for now: Slack is configured but nothing uses it yet, and the only email pattern in the system is vendor-facing. Either can be added later if needed.
- Alerts fire immediately, including after-hours — the emergency definition explicitly includes after-hours response.

Phase 4 — AI Receptionist Tie-In (Future)

- Expose the same classification logic as an internal API so the future AI Receptionist can ask "is this an emergency?" live during a tenant call and route accordingly (dispatch immediately vs. create a routine request).
- Keep the criteria in **one place** (a config the agent reads) so the dashboard and the receptionist can never disagree on what counts as an emergency.

7. Decisions

Question	Decision
Auto-set or suggest-only?	Fully automatic (per IT lead) — every new work order is labeled at intake with no review step. Ambiguous safety/damage cases lean Emergency. Manual staff settings are never overwritten and remain the correction path.
Who gets alerts, which channel?	In-app notification to all staff (existing notification bell) + SMS to the work order's assigned WOC (existing Twilio integration). Slack/email deferred.
Backfill existing work orders?	No — new work orders only (per IT lead). AI classification runs exactly once, when a work order first syncs in from PropertyWare. Once the emergency status is set — by AI or by staff — it is never updated again, and setting it auto-generates the work order's tasks.